📝 Chapter 2: Mechanics of Futures Markets (Continued)

3. Convergence of Futures Price to Spot Price (Section 2.3)

a. The Convergence Principle

- **General Rule:** As the delivery period for a futures contract approaches, the futures price converges to the spot price of the underlying asset.
- **At Delivery:** When the delivery period is reached, the futures price either **equals** or is **very close to** the spot price.

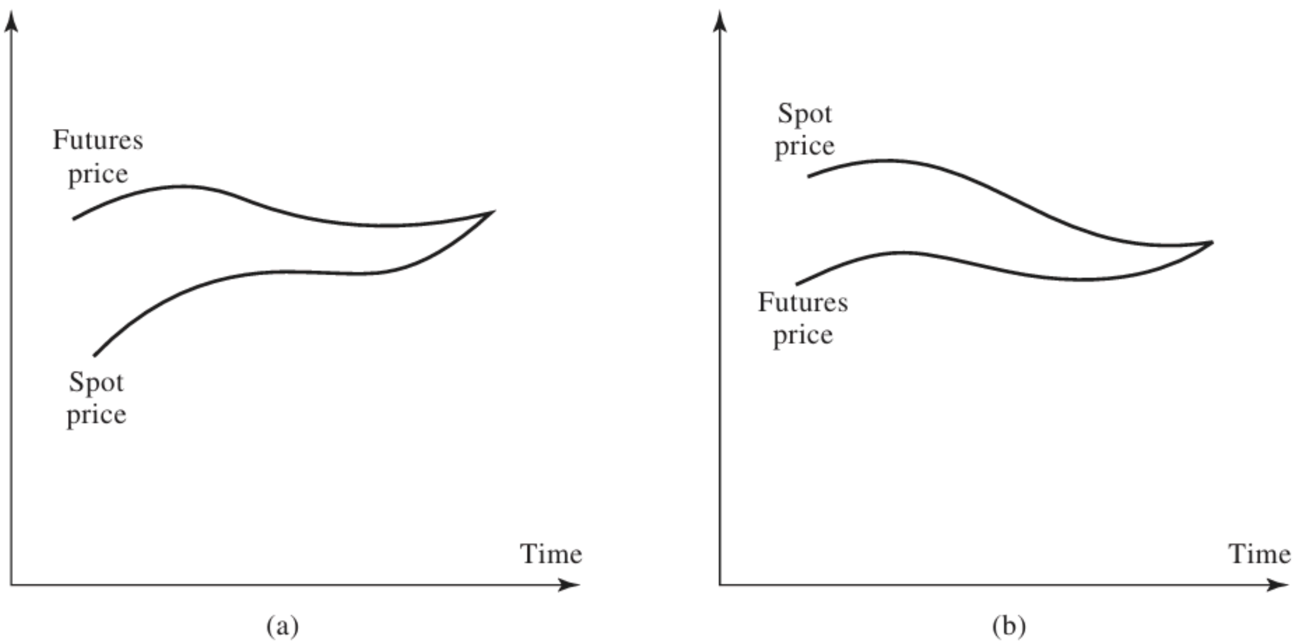
b. Arbitrage Forces Driving Convergence

The convergence is enforced by arbitrage opportunities:

Scenario	Arbitrage Action	Market Impact
Futures Price (F) > Spot Price (S)	1. Sell (short) a futures contract. 2. Buy the asset. 3. Make delivery.	This risk-free profit causes the **futures price (F) to fall**.
Futures Price (F) < Spot Price (S)	Companies acquire the asset by entering into a long futures contract and waiting for delivery.	This demand for long positions causes the **futures price (F) to rise**.

- **Result:** Arbitrage ensures that the futures price tracks the spot price closely up to the final day.

Figure 2.1 Relationship between futures price and spot price as the delivery period is approached: (a) Futures price above spot price; (b) futures price below spot price.



4. The Operation of Margin Accounts (Section 2.4)

a. Purpose of Margin Accounts

- **Risk Mitigation:** Margin accounts are the key mechanism used by the exchange to **avoid contract defaults** by ensuring traders have the financial resources to honor their agreements.

b. Daily Settlement (Marking to Market)

- **Initial Margin:** The amount of funds that must be deposited by the investor when the contract is entered into.
- **Daily Settlement:** At the end of each trading day, the margin account is adjusted to reflect the investor's gain or loss. This is also known as **marking to market**.
- **Flow of Funds:** Gains and losses are settled daily between brokers and the exchange clearing house, and money flows from the losing side's broker to the winning side's broker via the clearing house.
- **Effective Contract Rewriting:** A futures contract is, in effect, **closed out and rewritten at a new price each day** (the new settlement price).

c. Maintenance Margin and Margin Calls

- **Maintenance Margin:** A minimum account balance level, set lower than the initial margin, that must be maintained.
- **Margin Call:** If the margin account balance falls **below the maintenance margin**, the investor receives a margin call.
- **Variation Margin:** The extra funds deposited to bring the account balance back up to the **initial margin** level.
- **Default:** If the variation margin is not provided by the end of the next day, the broker **closes out the position** (neutralizes the contract).

d. Further Details on Margin

- **Interest:** Most brokers pay interest on the balance in the margin account.
- **Alternative Collateral (Initial Margin Only):** Securities can be deposited instead of cash to satisfy initial margin:
  - Treasury bills are accepted at about 90% of their face value.
  - Shares are sometimes accepted at about 50% of market value.
- **Setting Minimums:** Minimum margin levels are set by the exchange clearing house and are based on the **variability** of the underlying asset's price. The maintenance margin is usually about 75% of the initial margin.
- **Lower Requirements:**
  - **Bona Fide Hedgers** may face lower margin requirements due to less risk of default.
  - **Day trades** and **spread transactions** also often have lower requirements.
- **Symmetry:** Margin requirements are the **same for short futures positions as they are for long futures positions**.

e. The Clearing House and Credit Risk

- **Clearing House Role:** Acts as an **intermediary** in all transactions, guaranteeing the performance of the parties.
- **Clearing Margin:** Clearing house members (brokers) provide initial margin to the clearing house, often calculated on a **net basis** (short positions offset against long positions).
- **Guaranty Fund:** Members contribute to a fund used if a member fails to provide variation margin, ensuring traders' contracts are honored.
- **Success:** The margining system has been highly successful; contracts at major exchanges have always been honored.